

# Can a model that never saw your sales forecast them?

| Chronos-2 on M5 retail demand                                                                                  |                                                                  | Test before shipping                                                                  |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| From                                                                                                           | Shashwat Pratap Singh                                            |                                                                                       |
| To                                                                                                             | Anyone forecasting retail demand                                 |                                                                                       |
| Contents                                                                                                       | 1,000 Walmart item-store series, 6 models, one NVIDIA T4 (16 GB) |                                                                                       |
| Weight                                                                                                         | 120M parameters, zero tuning                                     |                                                                                       |
| Code and data<br><a href="https://github.com/Shrinet82/ML-Experiments">github.com/Shrinet82/ML-Experiments</a> |                                                                  |  |

# Best forecast, small edge.

Chronos-2 zero-shot against AutoETS, the best classical model

**-15%**

forecast error  
MASE 1.25 vs 1.46

**≈0**

inventory cost gain  
69.21 vs 69.59

**18×**

faster  
10s vs 190s

Lower MASE means a better forecast. The cost figure uses an assumed 3:1 stockout-to-overstock ratio.

**More accurate and much faster. In dollars it ties.**

# How I tested it

## Setup

# 1,000

M5 item-store series, random sample, seed 42

# 6

models: Chronos-2 twice, Bolt, AutoETS, seasonal naive, LightGBM

# 3 × 28

rolling origins, each a 28-day forecast

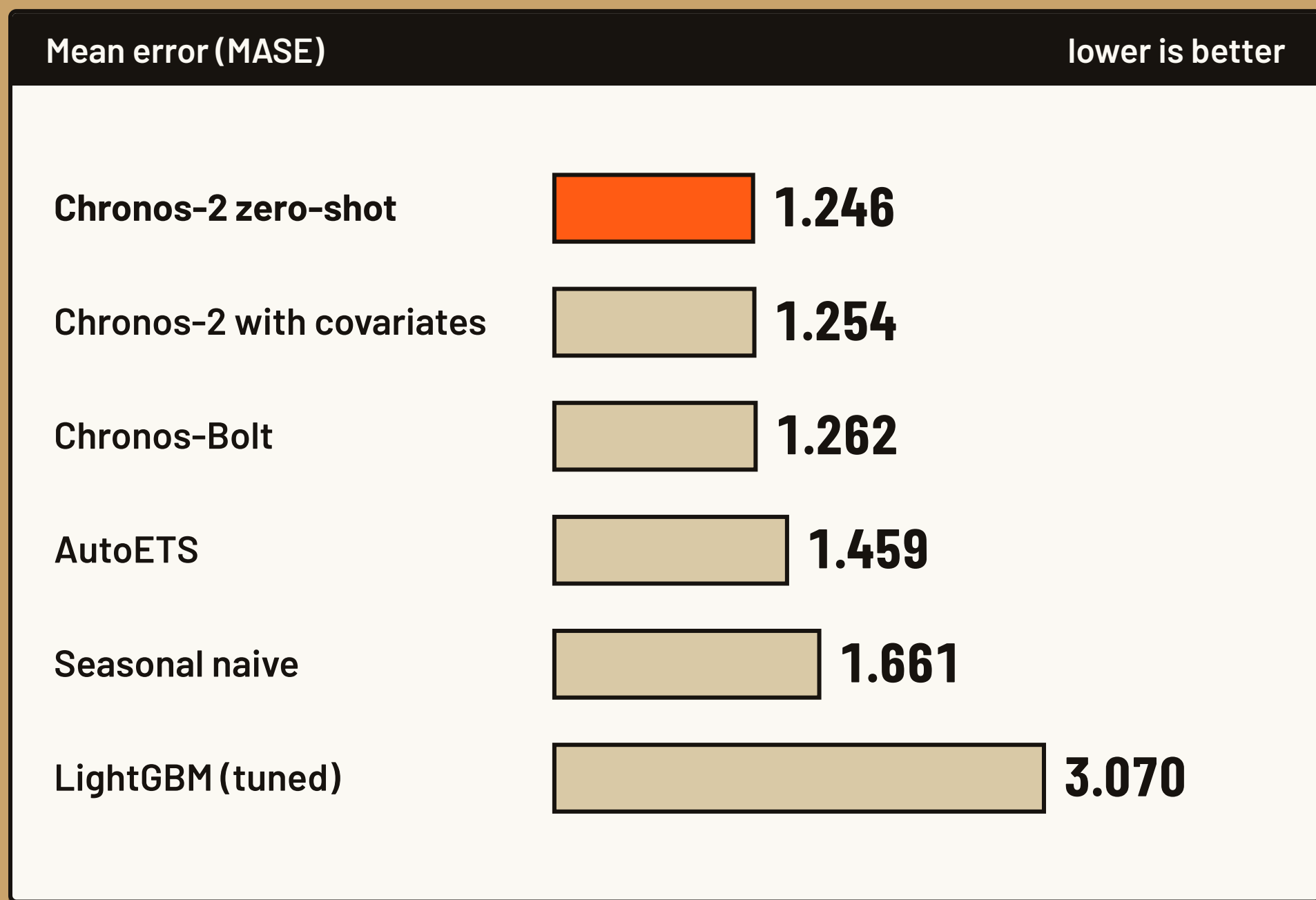
# 10k

paired bootstrap resamples over series

Chronos-2 also ran with price, event and SNAP covariates. TimesFM 2.5 was skipped, so it is not tested.

**The whole benchmark ran in 6 min 23 s on one T4.**

# All three Chronos models have the lowest error.



The three Chronos models sit within 0.016 of each other. LightGBM is a weak baseline here, worse than seasonal naive.

**Beating LightGBM proves little. Beating AutoETS counts.**

# Covariates did not pay off.

Error gap against each model, Chronos-2 with covariates minus baseline

Seasonal naive

**+0.408**

95% CI 0.379 to 0.437

Win

AutoETS

**+0.205**

95% CI 0.185 to 0.226

Win

Chronos-Bolt

**+0.008**

95% CI 0.001 to 0.015

Win

Chronos-2 zero-shot

**-0.008**

95% CI -0.013 to -0.002

Loss

Positive means the covariates version is better. Paired bootstrap over series, 10,000 resamples, no multiple-comparison correction. The Bolt gap is tiny.

**Adding price and event data made it slightly worse than zero-shot.**

# More data, worse forecast.

## Chronos-2 zero-shot, then with covariates

Error (MASE)

**1.246** → **1.254**

slightly worse

Quantile loss (WQL)

**5,452** → **12,734**

2.3× worse

Inventory cost at 3:1

**69.21** → **69.45**

no real change

Seconds per 1,000 series

**10** → **56**

5.3× slower

Why covariates did not help on M5 was not tested.

**Try zero-shot first. Covariates cost 5× the time here.**

# In dollars, it ties with AutoETS.

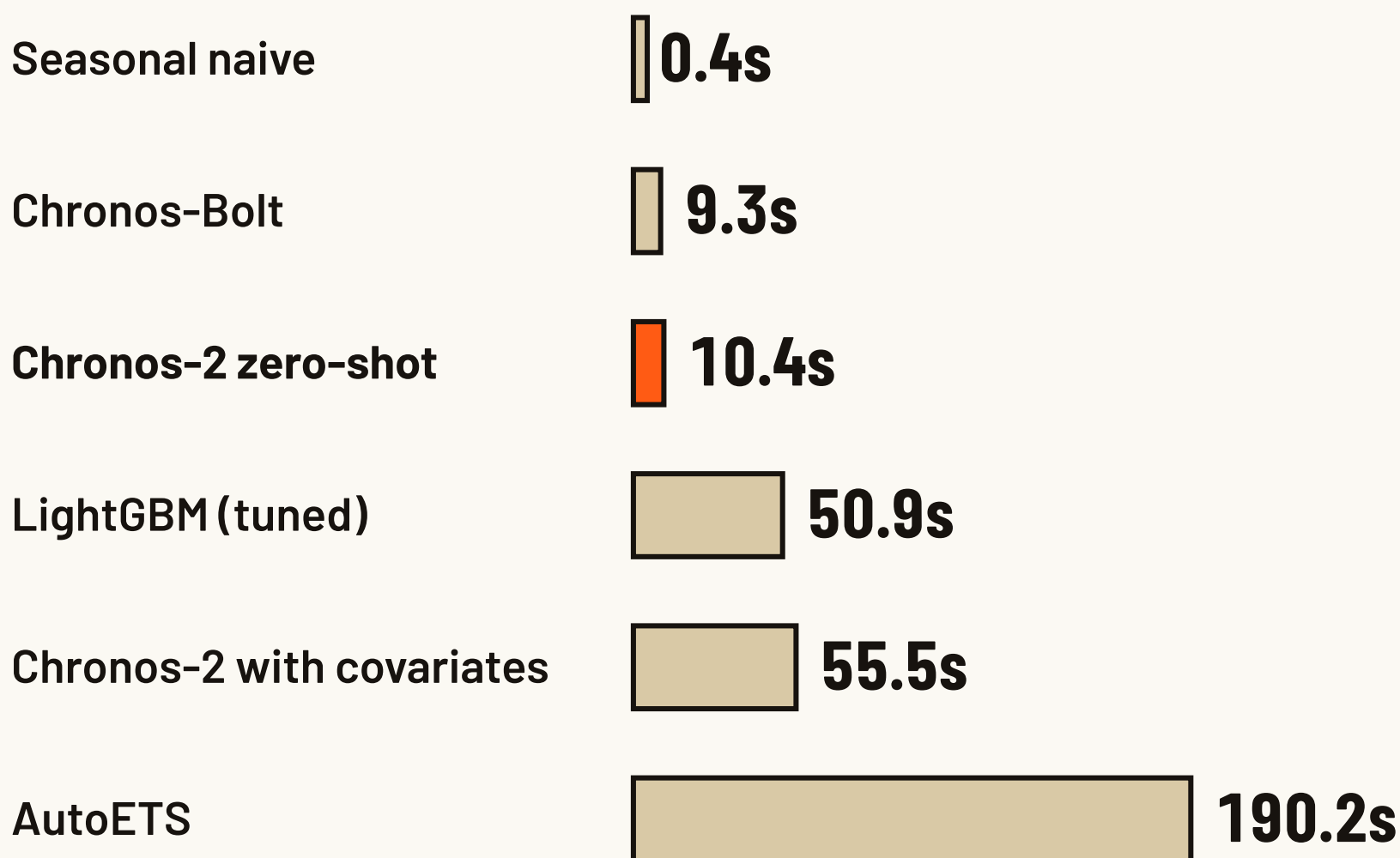
| Average newsvendor cost at the 0.9 quantile | stockout to overstock cost ratio |         |         |          |
|---------------------------------------------|----------------------------------|---------|---------|----------|
|                                             | 1:1                              | 3:1     | 5:1     | 9:1      |
| Chronos-2 zero-shot                         | \$60.04                          | \$69.21 | \$78.38 | \$96.71  |
| Chronos-2 with covariates                   | \$60.17                          | \$69.45 | \$78.72 | \$97.27  |
| Chronos-Bolt                                | \$59.38                          | \$69.30 | \$79.22 | \$99.07  |
| AutoETS                                     | \$60.71                          | \$69.59 | \$78.46 | \$96.21  |
| Seasonal naive                              | \$64.65                          | \$75.33 | \$86.01 | \$107.36 |

These cost ratios are my assumptions, not company data. Orange marks the lowest cost in each column.

Seasonal naive costs 9% more. Everything else is within noise.

# Seconds, not minutes.

Seconds per 1,000 series, three forecast origins



Chronos-2 zero-shot takes 10.4 seconds and AutoETS takes 190.2. Zero-shot needs no fitting at all.

**18× faster than AutoETS, on a single T4.**

# Most items are slow movers.

| Mean error (MASE) by demand type |  | lower is better |       |         |       |
|----------------------------------|--|-----------------|-------|---------|-------|
|                                  |  | Zero-shot       | Bolt  | AutoETS | Naive |
| Smooth, 232 series               |  | 0.820           | 0.838 | 0.845   | 1.043 |
| Intermittent, 768 series         |  | 1.374           | 1.390 | 1.644   | 1.848 |

Intermittent means more than half of the days had zero sales. The gap over AutoETS is biggest there, 1.37 against 1.64. In inventory cost AutoETS still ties: \$50.16 against \$51.02.

The biggest accuracy gain is on slow-moving items.

# Where to run it.

## Deployment options

### Laptop, CPU only

Supported by the model. Fine for development and small batches. I did not test CPU speed.

Not tested

### One GPU VM (T4, L4 or A10G)

The best fit for a nightly batch. My T4 run did about 288 forecasts per second. The model card quotes 300+ per second on an A10G.

Measured and model card

### AWS SageMaker with AutoGluon-Cloud

Managed endpoint with real-time, serverless and batch inference. The model card example uses ml.g5.xlarge.

Model card

### Your own container on a GPU node

Bake the weights into the image, pin versions and run it as a scheduled job.

My advice

**1M SKUs take about 58 minutes on one T4 (my estimate).**

# Ship it right.

## Checklist

**1****Run it as a nightly batch**

Store forecasts in a table. Don't serve per click.

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**2****Pin versions and bake in the weights**

chronos-forecasting 2.x, Apache-2.0 model.

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**3****Start zero-shot**

Covariates were 5× slower with no gain here.

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**4****Respect the model limits**

8,192 context, 1,024 horizon (model card).

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**5****Post-process the quantiles**

Crossing showed up in 1 to 4% of my runs. Sort them and clip below zero.

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**6****Monitor against seasonal naive**

If it stops beating naive, fall back and investigate.

**GPU, batch job, zero-shot. That is the setup.**

# A great default. Check the dollars.

## Use it when

**You have thousands of series**

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**You have no time to tune**

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**Your items sell slowly**

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**You need quantile forecasts**

## Don't expect

**Covariates to add lift**

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**A big dollar gain over AutoETS**

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**A fair LightGBM rival, mine was weak**

Limits: 1,000-series sample, one dataset, no multiple-comparison correction, quantile loss dominated by large series, assumed cost ratios, deployment speed measured on a T4 only, TimesFM not tested.

**Code and data**

**[github.com/Shrinet82/ML-Experiments](https://github.com/Shrinet82/ML-Experiments)**